

NWIS

WELL COMPLETION REPORT

Doc ID
W005-WCR-01

Well: Alpha-05 | Well ID: W005 | Field: Field_Alpha

PRIMARY ISSUE CATEGORY: TORQUE SPIKE

1. GENERAL WELL INFORMATION

Well Name	Alpha-05	Well ID	W005
Field	Field_Alpha	Block	Block_3
Spud Date	08-01-2025	Rig Release Date	07-02-2025
Total Depth (TD)	3400 m	Rig Name	Rig Sagar-5
Well Type	Directional Development	Trajectory	J-shaped (build 0-42 deg by 2500 m, hold to TD)
Max Inclination	42 deg	Directional Interval	2200-2500 m (build section)

2. FORMATION & PORE PRESSURE INFORMATION

Formation_W2 (interbedded shale and calcareous stringers) between 2700 m and 2800 m contains hard, abrasive calcareous stringers that have repeatedly caused bit whirl and torque instability in the tangent section of Block_3 directional wells, aggravated by a doglegs cluster around 2730-2750 m from the directional build.

3. LITHOLOGY SUMMARY

Interval (m)	Formation	Lithology	Notes
2400-2700	Formation_W1	Shale, grey, firm	Normal drilling trend
2700-2740	Formation_W2 (upper)	Shale with calcareous stringers	Torque instability zone
2740-2800	Formation_W2 (lower)	Calcareous stringer, hard/abrasive	Primary torque spike interval; dogleg cluster
2800-3100	Formation_X (upper)	Sand/shale interbedded	Normal drilling trend

4. HISTORICAL DRILLING / OPERATIONS EVENTS

Event 1: Torque Spike / Bit Whirl at approximately 2740 m

Observation: Surface torque spiked from a baseline of 10 kft-lb to 22 kft-lb within 3 minutes while drilling a hard calcareous stringer at the top of a dogleg cluster (build rate 4.2 deg/30m at 2735 m); downhole vibration data (if available) would be consistent with stick-slip and lateral bit whirl. RPM fluctuated erratically between 60-140 despite a constant surface setting of 100.

Engineering Action: WOB was reduced by 30% and RPM was reduced to 70 to break the stick-slip cycle. Torque-and-drag model was rerun in real time and compared against actual surface torque, confirming a friction factor increase consistent with a tight hole / key-seat forming through the dogleg cluster. A short wiper trip was performed back through the build section with a lubricant additive (2% ester-based lubricant) added to the mud system.

Outcome: Torque returned to a stable 11-13 kft-lb baseline after the wiper trip and lubricant treatment. ROP through the remaining stringer interval was reduced by approximately 35% as a precaution; 5.5 hours of NPT recorded.

5. CASING AND COMPLETION SUMMARY

Casing	Size (in)	Set Depth (m)	Cement Top (m)
Conductor	20	58	Surface
Surface	13-3/8	660	Surface
Intermediate	9-5/8	2850	1300
Production	7	3400	2450

6. NON-PRODUCTIVE TIME (NPT) SUMMARY

Issue	Depth (m)	NPT (hrs)	Category
Torque spike / wiper trip at 2740 m	2740	5.5	Torque & Drag / Hole Cleaning

7. ROOT CAUSE ANALYSIS

Root cause was a combination of hard, abrasive calcareous stringers within Formation_W2 and a concentrated dogleg cluster (4.2 deg/30m) from the directional build, which together elevated the friction factor along the BHA and created localized key-seating. The actual torque-and-drag trend diverged from the pre-well model by more than 25% through this specific interval, indicating the model's assumed friction factor (0.25) was too low for this stringer/dogleg combination.

8. LESSONS LEARNED

- Calcareous stringers in Formation_W2 (2700-2800 m) combined with dogleg severity above 4 deg/30m reliably produce torque spikes in Block_3 directional wells.
- Reducing WOB by 25-30% and RPM to 65-75 is an effective first response to break a stick-slip cycle before it escalates to a full torque lock-up.
- Real-time torque-and-drag model comparison (actual vs. modeled) identified the friction factor mismatch quickly; recommend updating the block-wide friction factor assumption from 0.25 to 0.32 for this stringer interval.
- 2% ester-based lubricant additive measurably reduced torque after the wiper trip; keep this on standby for any well planned through Formation_W2.

9. RECOMMENDATION FOR OFFSET WELLS

For offset directional wells in Block_3, update the pre-well torque-and-drag model friction factor to 0.32 for the 2700-2800 m interval, plan dogleg severity below 3.5 deg/30m through the stringer zone where feasible, and pre-stage ester-based lubricant additive for immediate use if torque exceeds 1.5x baseline.

This Well Completion Report is a synthetically generated historical record produced for the NWIS prototype (W005). Not associated with an actual drilling operation.